

Explore the influencing factors of education expenditure in each region

Abstract

This study employs econometric methods to explore the factors influencing educational expenditure in Chinese cities. Using software like SPSS and Stata, regression analysis was conducted on several independent variables, including city GDP, natural growth rate, government expenditure, the number of students in compulsory education, total student numbers, educational level, and residents' disposable income. Preliminary results showed heteroskedasticity in the model, leading to the decision to log-transform the dependent variable of educational expenditure in each city and develop a new regression model.

The revised model underwent an economic significance test, including a detailed interpretation of the regression results. Statistical significance tests included goodness-of-fit, significance judgment, t-tests, and F-tests. Further, this study conducted econometric tests such as multicollinearity (using the Variance Inflation Factor, VIF), heteroskedasticity, and autocorrelation tests to ensure the robustness of the model.

The final results revealed that GDP, natural growth rate, education level, the number of students in compulsory education, and disposable income of residents positively correlate with urban educational expenditure. Conversely, the total number of students and local government fiscal spending negatively impact educational expenditure.

Keywords: Educational expenditure; Multiple regression analysis; Model test

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1. Introduction

1.1 Background Analysis

Educational expenditure is a crucial fiscal investment for the development of educational undertakings by nations and regions, directly related to the quality and equity of education, as well as the long-term social and economic development of a region. With the deepening of globalization and the rise of the knowledge economy, education has become an important indicator of national competitiveness (Zhu&Yu, 2023). Therefore, exploring the factors affecting educational expenditure in various regions is of great significance for optimizing resource allocation, enhancing educational efficiency, and implementing educational reform directions.

The disparities in educational expenditure reflect the uneven levels of economic development between regions. Developed areas, due to their strong economic strength, usually can invest more resources in education, while less economically developed areas may struggle to provide sufficient educational funding due to fiscal constraints. Additionally, variations in government policies also affect educational spending. The choices made by different regional governments regarding educational priorities, funding allocation mechanisms, and tax policies will directly influence the scale and pattern of investment in education (Tang, Sun & Yang, 2021).

Differences in population structure are also a key factor affecting educational expenditures. For instance, regions with a higher proportion of students or more schools might require more educational resources to meet the needs of different school-age populations. Moreover, urban-rural structure, societal demands, and cultural backgrounds can also impact educational spending. In communities or cities that value education, there may be a greater inclination to invest in education, further developing the educational sector.

1.2 Research Objectives

The main purpose of this study is to explore and find out the key factors affecting the education expenditure in each region. This paper analyzes the impact of economic development level, government policy, population structure and other factors on education investment, hoping to provide more effective suggestions for the allocation of education funds and education reform. This not only helps to improve the efficiency of the use of existing resources, but also has important significance for promoting educational equity and improving the quality of education.

The study also aims to provide data support and a decision-making basis for policymakers, assisting them in the rational allocation of educational budgets in the face of limited fiscal resources, especially in regions where resources are relatively scarce. Furthermore, through analysis, the study aims to reveal the best practices and potential issues in the educational expenditures of different regions, offering references for the allocation of educational resources.

2. Data Description

2.1 Data Collection

Data related to the research question were collated from China Statistical Yearbook 2021, China Educational Expenditure Statistical Yearbook 2021, China Economic Private Network and National Bureau of Statistics of the People's Republic of China, as shown below:

Table 1 Collected variables

Number	Name	Shortened Form
1	log(Education Expenditure)	ledu
2	Education expenditure (10 thousand CNY)	edu
3	Location	L
4	Gross Domestic Product (GDP) (10 thousand CNY)	GDP
5	GDP per capita (CNY)	CGDP
6	Natural Growth Rate (‰)	NGR
7	Average Salary of Employees (CNY)	ASE
8	Expenditure within the general budget of local finance (10 thousand CNY)	EXP
9	Education Expenditure/Fiscal Expenditure (%)	EE
10	Science Expenditure (10 thousand CNY)	SE
11	Number of Students in Regular Institutions of Higher Education (Person)	SRIH
12	Number of students enrolled in undergraduate and junior college programs (Person)	SEUJ
13	Number of students enrolled in regular secondary schools (ten thousand Person)	SERS
14	Number of students enrolled in primary school (ten thousand Person)	SEPS
15	Number of primary school students (ten thousand Person)	PST
16	Number of Students Enrolled in Secondary Vocational and Technical Schools (Person)	SVT
17	Number of Students Enrolled in Adult Higher Education Institutions (Person)	SEAH
18	Number of Students receiving compulsory education (Person)	SCE
19	Number of students	S
20	Educational level (tertiary)	EDUL
21	Disposable income of residents	INC
22	CPI (2021)	CPI

2.2 Variable Selection

Education expenditure is taken as the dependent variable and data 3-22 as the independent variable. spss was used for multiple regression analysis. According to the test results, 7 independent variables were selected (Selected on the basis of significance and collinearity), which were as follows: Gross Domestic Product (GDP), Natural Growth Rate, Expenditure within the general budget of local finance, Number of Students receiving compulsory education, Number of students , Educational level and Disposable income of residents.

3. Establishment of model

This paper adopts multiple regression model (Model 1)

$$Y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + \beta_4x_4 + \beta_5x_5 + \beta_6x_6 + \beta_7 + u$$

Y: Education expenditure

x_1 : Gross Domestic Product (GDP)

x_2 : Natural Growth Rate

x_3 : Expenditure within the general budget of local finance

x_4 : Number of Students receiving compulsory education

x_5 : Number of students

x_6 : Educational level

x_7 : Disposable income of residents

u: Error term

β_0 : Intercept

$\beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6, \beta_7$: Slope parameters

4. Optimization of model

4.1 Model solving and analysis

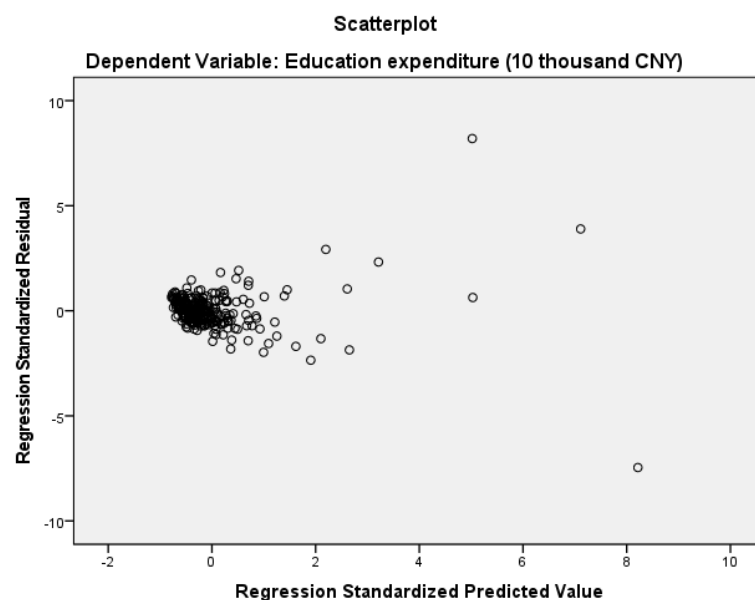
The established model was estimated using the ordinary least square method, and relevant tests and corrections were made to the results obtained. The regression results are shown in the table below, from which it can be found that some independent variables have high significance, indicating that the selected independent variables cannot explain the changes of dependent variables well.

Table 2 Coefficient of multiple regression in Model 1

Coefficients ^a											
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Correlations			Collinearity Statistics	
		B	Std. Error	Beta			Zero-order	Partial	Part	Tolerance	VIF
1	(Constant)	-116849.055	81892.739		-1.427	.155					
	Gross Domestic Product (GDP)	.007	.001	.294	6.496	.000	.955	.377	.067	.106	9.473
	Natural Growth Rate	21777.596	3669.106	.068	5.935	.000	.156	.348	.061	.799	1.252
	Expenditure within the general budget of local finance	.105	.006	.662	18.378	.000	.974	.755	.189	.136	7.327
	Number of Students receiving compulsory education	.407	.142	.142	2.867	.004	.675	.177	.029	.123	8.152
	Number of students	-.098	.108	-.068	-.913	.362	.729	-.057	-.009	.100	9.989
	Educational level	-.139	.235	-.021	-.589	.556	.544	-.037	-.006	.148	6.756
	Disposable income of residents	.312	1.928	.003	.162	.871	.566	.010	.002	.424	2.358
	a. Dependent Variable: Education expenditure (10 thousand CNY)										

Then, it can be seen from the scatter plot of standardized residuals and standardized predicted values that the dispersion fluctuation range of standardized residuals spreads to the right, so the model has obvious heteroscedasticity.

Figure 1 Residual scatter plot



Therefore, we further optimize the model. Logarithmic transformation of the model can reduce the difference between the values and obtain a model with higher fitting degree.

4.2 Model optimization process

The dependent variable Education expenditure (10 thousand CNY) of the above model is transformed logarithmically to obtain the data $\log(\text{Education Expenditure})$, which is used as the new dependent variable. Education expenditure, GDP per capita, Natural Growth Rate, Average Salary of Employees, Expenditure within the general budget of local finance, Number of students enrolled, Educational level and Disposable income of residents as independent variables, establish a new multiple regression model (Model 2)

$$\log(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 + u$$

5. Model solving and analysis

5.1 Model solving

We use SPSS to estimate the established model by using ordinary least square method, and perform relevant tests on the obtained results. The results are as follows:

Table 2 Coefficient of multiple regression in Model 2

Coefficients ^a											
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Correlations			Collinearity Statistics	
		B	Std. Error	Beta			Zero-order	Partial	Part	Tolerance	VIF
2	(Constant)	5.222	.056		93.536	.000					
	Gross Domestic Product (GDP)	1.955E-9	.000	.312	2.609	.010	.777	.161	.071	.106	9.473
	Natural Growth Rate	.006	.003	.072	2.382	.018	.299	.148	.065	.799	1.252
	Expenditure within the general budget of local finance	-1.238E-9	.000	-.030	-.318	.024	.724	-.020	-.009	.136	7.327
	Number of Students receiving compulsory education	5.435E-7	.000	.732	5.615	.000	.811	.332	.152	.123	8.152
	Number of students	-1.045E-7	.000	-.278	-1.424	.016	.814	-.089	-.039	.100	9.989
	Educational level	2.899E-7	.000	.168	1.805	.002	.581	.112	.049	.148	6.756
	Disposable income of residents	6.489E-6	.000	.205	4.938	.000	.520	.295	.134	.424	2.358
a. Dependent Variable: log(Education Expenditure)											

5.2 Economic significance test

In economic terms, the investment in education funding in various regions should have a linear relationship with Gross Domestic Product (GDP), Natural Growth Rate, Expenditure within the general budget of local finance, Number of Students receiving compulsory education, Number of students, Educational level, and Disposable income of residents. According to the OLS regression results, it can be observed that $\beta_0, \beta_1, \beta_2, \beta_4, \beta_6, \beta_7$ are greater than 0, and β_3, β_5 are less than 0.

In model2, β_1 represents that for every unit increase in Gross Domestic Product, there will be a corresponding increase of 1.955E-9 units in the log(education Expenditure). $\beta_2, \beta_4, \beta_6, \beta_7$ also represents the corresponding marginal utility.

β_3 indicates that for every unit increase in Expenditure within the general budget of local finance, there will be a corresponding decrease of 1.238E-9 units in the log(education funding). β_5 also represents the corresponding marginal utility.

The negative value of β_3, β_5 can be understood as when the government allocates financial resources, it may focus on other areas. If education expenditure is relatively low, there will be a negative correlation between general budget expenditure and education expenditure. Similarly, when the number of students increases but the local government's budget for education does not increase, there will be a negative correlation between the number of students and education. Therefore, β_3, β_5 is considered reasonable and meaningful.

5.3 Statistical significance test

5.3.1 Goodness-of-fit test

The regression analysis results are as follows:

Table 3 Model 2 Summary

Model Summary ^b									
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
2	.902 ^a	.813	.808	.15408788654 48	.813	158.445	7	255	.000

a. Predictors: (Constant), Disposable income of residents, Natural Growth Rate (%), Educational level, Number of Students receiving compulsory education, Expenditure within the general budget of local finance, Gross Domestic Product (GDP), Number of students
b. Dependent Variable: log(Education Expenditure)

➤ **The coefficient of determination (R-squared)**

R-squared is a commonly used indicator to evaluate the goodness of fit of a model. It represents the proportion of the total variation in the dependent variable that can be explained by the regression relationship with the independent variables. The value of R-squared ranges from 0 to 1, where a value closer to 1 indicates a better fit of the model to the observed data. Since R-squared tends to increase with the number of independent variables, adjusted R^2 is used to account for the number of independent variables and provide a more accurate assessment of the model's fit.

In this model, $R^2 = 0.813$ and Adjusted $R^2 = 0.808$, indicating that the regression relationship explains over 80% of the variation in the dependent variable and suggesting a good fit of the model to the observed data.

➤ **Standard error(σ)**

Standard error(σ) is a measure of the difference between the sample mean and the population mean. A smaller standard error indicates higher precision of the estimated values.

In this model, $\sigma = 0.154087886544818$, indicating that the fitted values are sufficiently close to the observed values.

5.3.2 Significance test

➤ **T-test (The result in 5.1 Table 2)**

The t-statistic and p-value are used to assess the statistical significance of each independent variable. The t-statistic measures the ratio of the coefficient of the independent variable to its standard error, while the p-value indicates whether the coefficient is significantly different from zero in the population. Typically, a p-value less than 0.05 is considered statistically significant. In this model, all the p-values for the independent variables are less than 0.025, indicating that the effects of these variables on education expenditure are statistically significant.

➤ **F-test (The result in 5.1 Table 2)**

The F-statistic is used to test the overall significance of the model's fit. In this model, the F-value is 158.445, and its corresponding p-value is less than 0.05. This indicates that the overall fit of the

model is significant, and the overall effect of the model is statistically significant in the population. It suggests that the model provides a good overall fit to the data.

5.4 Econometric Significance Tests

5.4.1 Collinearity Test

Multicollinearity refers to the high correlation between two or more explanatory variables in a regression model. This correlation can lead to instability in the estimation of regression coefficients and an increase in standard errors, thereby affecting the explanatory power and predictive accuracy of the model. The Variance Inflation Factor (VIF) is the most commonly used quantitative metric for detecting multicollinearity. It measures how the variance of an explanatory variable is inflated due to its linear relationship with other variables.

VIF Test

$$VIF_i = \frac{1}{1 - R_i^2}, i = 1, 2, 3, 4, \dots, k$$

R_i^2 : The coefficient of determination obtained from the linear regression of explanatory variables. Through the above model analysis and selection of correlation analysis data, the following table is obtained:

Table 4 The result of VIF test

Model		Collinearity Statistics	
		Tolerance	VIF
2	(Constant)		
	Gross Domestic Product (GDP)	.106	9.473
	Natural Growth Rate (‰)	.799	1.252
	Expenditure within the general budget of local finance	.136	7.327
	Number of Students receiving compulsory education	.123	8.152
	Number of students	.100	9.989
	Educational level	.148	6.756
	Disposable income of residents	.424	2.358
Dependent Variable: log(Education Expenditure)			

From the table above, it can be seen that the VIF values of each independent variable are all less than the threshold of 10, indicating that there is no multicollinearity in this regression.

5.4.2 Heteroskedasticity Diagnosis

In econometrics, the significance of heteroskedasticity tests lies in determining whether the error terms of a regression model have constant variance. If this condition is not met, indicating the presence of heteroskedasticity, the standard errors of regression analysis might be incorrectly estimated, thus affecting the statistical significance tests of coefficient estimates and the accuracy of confidence intervals.

Common methods for testing heteroskedasticity include the Breusch-Pagan test and the White test.

These tests determine whether heteroskedasticity still exists in the optimized model.

➤ Breusch-Pagan test

Conducting the Breusch-Pagan test for the selected dependent and independent variables:

I. Setting the null hypothesis

H_0 : NO Heteroskedasticity

II. Obtaining data for the Breusch-Pagan Test and Modified Breusch-Pagan Test.

Table 5 The result of Breusch-Pagan Test

Breusch-Pagan Test for Heteroskedasticity^{a,b,c}		
Chi-Square 1427.979	df 1	Sig. .000
a. Dependent variable: ledu		
b. Tests the null hypothesis that the variance of the errors does not depend on the values of the independent variables.		
c. Predicted values from design: Intercept + GDP + NGR + EXP + SCE + S + EDUL + diru		

Table 6 The result of Modified Breusch-Pagan Test

Modified Breusch-Pagan Test for Heteroskedasticity^{a,b,c}		
Chi-Square 10.952	df 1	Sig. .001
a. Dependent variable: ledu		
b. Tests the null hypothesis that the variance of the errors does not depend on the values of the independent variables.		
c. Predicted values from design: Intercept + GDP + NGR + EXP + SCE + S + EDUL + diru		

III. Conclusions

As indicated in the table above, due to the excessively large chi-square value of the BP-test and its significance level being extremely low, approaching zero, we analyze the data from the modified BP-test. At the 0.01 significance level, the Breusch-Pagan test shows a chi-square value greater than

the degrees of freedom ($10.952 > 1$), indicating insufficient evidence to reject H_0 , meaning the data does not exhibit heteroskedasticity.

➤ **White Test**

The White test is a method for detecting heteroskedasticity in regression models. Its approach is similar to the Breusch-Pagan test, but it does not require specific assumptions about the form of error variance. In the White test, the chi-square statistic is also used to determine whether to reject the null hypothesis that the error terms of the regression model have constant variance. This statistic is compared with the critical value of the chi-square distribution for the corresponding degrees of freedom:

I. Setting the null hypothesis

H_0 : NO Heteroskedasticity

II. Obtaining data for the White Test

Table 6 The result of White Test

White Test for Heteroskedasticity^{a,b}		
Chi-Square 261.000	df 262	Sig. .471
a. Dependent variable: ledu		
b. Tests the null hypothesis that the variance of the errors does not depend on the values of the independent variables.		

II. Conclusions

As shown in the table above, the chi-square statistic of the White test is less than the critical value of the chi-square distribution for the given degrees of freedom and significance level. Therefore, we do not have sufficient evidence to reject the null hypothesis, suggesting that the error terms of the model have constant variance, meaning there is no heteroskedasticity present.

5.4.3 Autocorrelation Test

The autocorrelation test is a method in statistical analysis used to determine whether there is a correlation between consecutive observations in time series data. In regression analysis, our concern is whether the residuals (the error terms of the regression model) exhibit autocorrelation, as this could violate one of the fundamental assumptions of the classical linear regression model, which is that the error terms are independent of each other.

The Durbin-Wassen test was used here and the results are shown below:

Table 7 The result of Durbin-Watson Test

Model	R	R Square	Durbin-Watson
2	.902a	.813	1.770

Its values typically range from 0 to 4, where a value close to 2 indicates no autocorrelation, close to 0 indicates positive autocorrelation, and close to 4 indicates negative autocorrelation.

As shown in the table above, the D-U value is 1.770, close to 2 and relatively far from 0, indicating that there is no autocorrelation in this regression.

6. Summary

The final research results reveal that Gross Domestic Product (GDP), Natural Growth Rate, Expenditure within the general budget of local finance, Number of Students receiving compulsory education, Number of students, Educational level, and Disposable income of residents have a linear relationship with government expenditure in the education sector. The obtained model is as follows:

$$\log(\text{edu}) = 5.222 + (1.955\text{E} - 9)\text{GDP} + 0.006\text{NGR} + (-1.238\text{E} - 9)\text{EXP} + (5.435\text{E} - 7)\text{SCE} \\ + (-1.045\text{E} - 7)\text{S} + (2.899\text{E} - 7)\text{EDUL} + (6.489\text{E} - 6)\text{INC} + u$$

The model has passed the tests of economic significance, statistical inference, and econometric analysis, thereby validating the reliability and effectiveness of our conclusions in the field of economics. Based on this model, we can provide reasonable and effective recommendations for policy formulation and decision-making in the government's education sector. We can also make projections about government expenditure in the education sector based on future relevant variables.

The field of education is a solid foundation for promoting knowledge development and talent cultivation in our country, as well as driving improvements in productivity and comprehensive national strength. Therefore, research in the education sector is highly important and valuable, including the examination of government expenditure in education. We hope that education in our country can reach new heights to achieve the strategic goal of revitalizing the nation through science and education.

References

- Draper, N.R. & Smith, H. (1998). *Applied Regression Analysis*. *Wiley Series in Probability and Statistics*.
- Farayibi, A. O., & Folarin, O. (2021). Does government education expenditure affect educational outcomes? New evidence from sub-Saharan African countries. *African Development Review*, 33(3), 546–559. <https://doi.org/10.1111/1467-8268.12588>
- Haini, H. (2020). Spatial spillover effects of public health and education expenditures on economic growth: evidence from China's provinces. *Post-Communist Economies*, 32(8), 1111–1128. <https://doi.org/10.1080/14631377.2020.1722586>
- Scientific Platform Serving for Statistics Professional. (2021). SPSSPRO (Version 1.0.11) [Online Application Software]. Retrieved from <https://www.spsspro.com>
- Tang, L., Sun, S., & Yang, W. (2021). Does government education expenditure boost intergenerational mobility? Evidence from China. *International Review of Economics & Finance*, 74, 13–22. <https://doi.org/10.1016/j.iref.2021.02.001>
- Zhu, Y., & Yu, D. (2023). Education and happiness: does education expenditure undermine households' subjective well-being? *Evidence from China*. *Applied Economics*, 55(50), 5925–5938. <https://doi.org/10.1080/00036846.2022.2140772>